

RENANCE

PETROLEUM TRAINING INSTITUTE, EFFURUN

MEC 112 — Past Questions & Materials

by Renance

Course: MEC 112

Institution: Petroleum Training Institute, Effurun

Papers in this pack: 2

Levels: ND1

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Paper 1 of 2 — MEC 112 - Mechanical Engineering Science I: Answers and Resources

Level ND1 • First Semester • Handout

MEC 112 Exam Answers with YouTube Links

Q1

Explain the principle of moment:

The moment of a force is the product of the force and the perpendicular distance from the point to the line of action

Q2

Find the forces in the members of the truss using the method of joints.

Q6

Simple frame analysis and tension/compression analysis.

YouTube Learning Links:

Vector Diagram Method: <https://www.youtube.com/watch?v=NjJf6KEoOjY>

Bracket Force Analysis: https://www.youtube.com/watch?v=Wli_1evfZ3Q

Simple Frame Analysis: <https://www.youtube.com/watch?v=DRRgmULUxhM>

Crank Lever Force: <https://www.youtube.com/watch?v=m0QpeP91bnM>

Paper 2 of 2 — MEC 112 - Mechanical Engineering Science I: Answers and Resources

Level ND1 • First Semester • Handout

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QUESTION 1

QUESTION 1

Answer:

(a) Principle of Moment:

The principle of moment states that when a body is in equilibrium, the sum of the clockwise moments about a point is equal to the sum of the anticlockwise moments about that same point.

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(b) Force Resolution:

Given force $F = 100 \text{ N}$, $\angle CAB = 30^\circ$, $\angle CBA = 15^\circ$.

Using triangle of forces and sine rule:

$$F_{AC}/\sin(15^\circ) = F_{BC}/\sin(30^\circ) = 100/\sin(135^\circ)$$

$$\rightarrow F_{AC} \approx 51.76 \text{ N}, F_{BC} \approx 25.88 \text{ N}$$

(c) Tension in ropes:

Using Lami's Theorem for 3-force equilibrium, apply the angles and known force to determine tensions in ropes A and B. Diagram is provided.

QUESTION 2

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QUESTION 2

Answer:

(a)

(i) Scalar: Quantity with magnitude only (e.g., speed, mass)

Vector: Quantity with magnitude and direction (e.g., velocity, force)

(ii) Coplanar: Forces in same plane

Non-coplanar: Forces not in same plane

(iii) Resultant: Single force equivalent to all acting forces

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Equilibrant: Force that balances out the resultant to keep system in equilibrium

(b) Conditions of Equilibrium:

- $\sum F_x = 0$, $\sum F_y = 0$ (no net force)

- $\sum \text{Moment} = 0$ (no net rotation)

(c) Draw FBD by isolating a joint, showing all connected forces.

(d) Method of Joints: Start from known support and solve joint-by-joint. EF = tension, FD = compression, DB = compression (sample values calculated step-by-step).

QUESTION 3

QUESTION 3

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Answer:

(a) Newton's Laws:

1st: A body remains at rest or in uniform motion unless acted upon.

2nd: $F = ma$

3rd: Every action has an equal and opposite reaction

(b)

(i) Rod weight = 5 N at center. Moments taken about hinge to find vertical and horizontal reaction components.

→ $R_{Ax} = 0$ N, $R_{Ay} \approx 15$ N

(ii) Diagram shown with hinge, forces, reactions, and rod center of gravity.

QUESTION 4

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QUESTION 4

Answer:

(a) Friction is a resistive force that opposes motion between surfaces in contact.

(b)

- Static: Prevents motion (e.g., parked car on slope)

- Dynamic: Opposes motion once in progress (e.g., sliding furniture)

Applications:

Static: Bolted joints, ladder placement

Dynamic: Braking system, conveyor belt

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(c) Coefficient $\mu = F/N = 100/100 = 1.0$

QUESTION 5

QUESTION 5

Answer:

(a) Triangle law: Place vectors head-to-tail; draw resultant from start to end.

Parallelogram: Draw both vectors from origin; diagonal is the resultant.

Forces $F_1 = 60 \text{ N}$, $F_2 = 80 \text{ N}$, $\theta = 60^\circ$

$\rightarrow R = \sqrt{(F_1^2 + F_2^2 + 2F_1F_2\cos\theta)} = \sqrt{(60^2 + 80^2 + 2 \times 60 \times 80 \times \cos 60^\circ)} \approx 123.69 \text{ N}$

(b)

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(i) Use equilibrium equations at supports to find R

(ii) Method of joints: resolve forces at nodes. Member AD = tension or compression based on direction.

(c) Methods:

- Method of Joints

- Method of Sections

QUESTION 6

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QUESTION 6

Answer:

(a)

Load = 2400 kg $\rightarrow$ 23544 N

Using moment about A and $\sum F_y$:

$\rightarrow R_B = 17,196 \text{ N}, R_{Ay} = 12,348 \text{ N}, R_{Ax} = 0 \text{ N}$

(b) Couple is used to produce pure rotation, e.g., tightening a bolt with a spanner.

(c) Resolve angled force into vertical and horizontal, apply equilibrium to solve pivot reaction.

Diagram shown.

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